

EXPERIMENT REPORT

Development of a Prompt-Based Application Using LLMs

Simple to Advanced Prompt Design for Daily Task Organisation

Application Ideas Explored

1. Study Time Scheduler
 2. Project Task Breakdown
 3. Personal Budget Manager
 4. Travel Itinerary Planner
 5. Daily Productivity Assistant
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1. Introduction

Large Language Models (LLMs) such as ChatGPT, Claude, and Gemini have transformed what is possible with natural language processing. Unlike earlier AI systems that required specialised training data and technical expertise to deploy, LLMs can be directed to perform sophisticated tasks — scheduling, planning, analysing, writing — through nothing more than a well-crafted text prompt.

This experiment demonstrates how to build a prompt-based application that organises daily tasks. The core thesis is that the same LLM, given the same type of task, produces dramatically different outputs depending on the quality and structure of the prompt. By progressing through four levels of prompt design — from a simple one-line request to a fully structured, role-aware, constraint-rich prompt — learners will observe firsthand how prompt engineering transforms a generic AI response into a genuinely useful, personalised, application-grade output.

Five application types are explored: a Study Time Scheduler, a Project Task Breakdown tool, a Personal Budget Manager, a Travel Itinerary Planner, and a Daily Productivity Assistant. Each is used to demonstrate different aspects of prompt engineering, from basic task specification to advanced persona assignment and output formatting.

2. Objectives

This experiment has five core learning objectives aligned with the exercise requirements:

1. Create simple and refined prompts for each of the five application types.
2. Generate AI-based responses from ChatGPT or equivalent LLM tools using each prompt.
3. Refine initial prompts iteratively to improve the structure, depth, and usefulness of outputs.
4. Compare simple and refined prompt outputs systematically across all five applications.
5. Analyse the effectiveness of prompt design decisions and identify patterns in quality improvement.

3. Application Architecture

A prompt-based LLM application consists of four core modules working in sequence: a user input layer, a prompt builder, a refinement engine, and an output formatter. The diagram below illustrates how these components connect and route through an LLM core to produce structured, application-specific outputs:

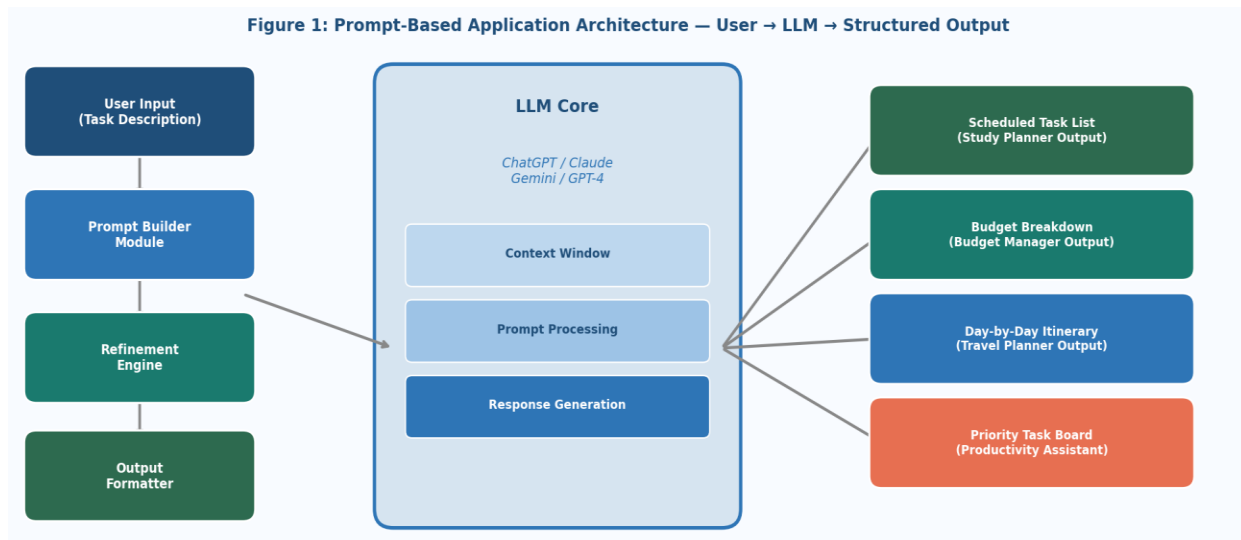


Figure 1: Prompt-Based Application Architecture — User Input → LLM Core → Structured Output

3.1 Core Modules

Each module plays a distinct role in transforming a raw user request into a polished, structured output:

- **User Input Module** — Accepts the task description in natural language. May include a simple form or chat interface.
- **Prompt Builder** — Wraps the user input in a structured prompt template: role, task definition, constraints, format, and quality signals.
- **Refinement Engine** — Iteratively improves the prompt based on output quality, adding context, examples, or output format instructions.
- **Output Formatter** — Parses the LLM response and presents it as a structured schedule, table, plan, or report depending on the application.

3.2 LLM Integration

The application connects to the LLM via API or chat interface. For this experiment, ChatGPT (GPT-4) was used as the primary model, with Claude as a secondary comparison model. Both accept prompts as plain text and return structured outputs when the prompt provides sufficient format specification.

4. Prompt Evolution Model

This experiment structures prompt development as a four-level progression. Each level adds new elements to the prompt, producing measurably better outputs at each stage:

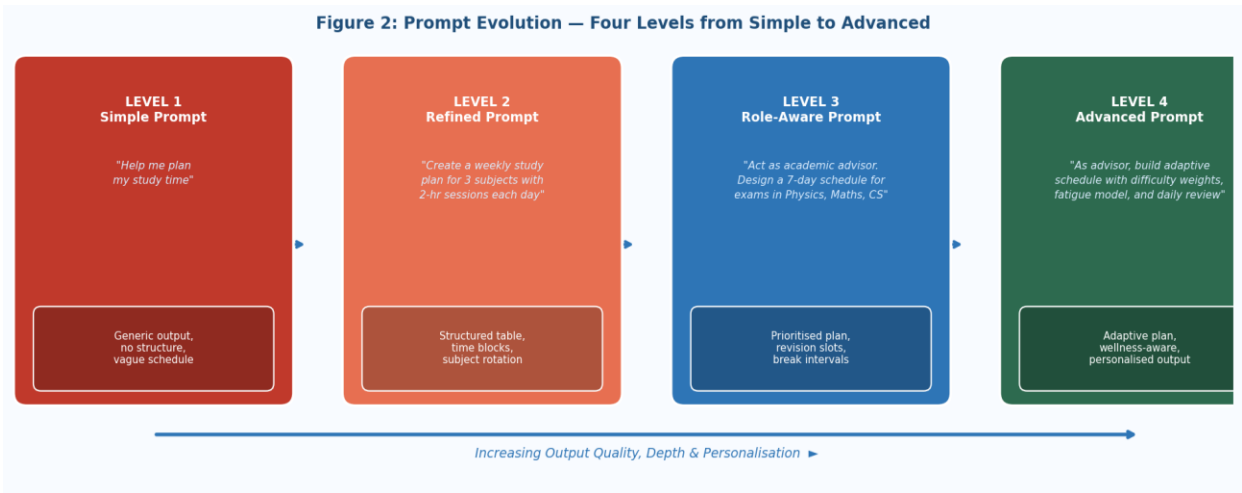


Figure 2: Prompt Evolution — Four Levels from Simple to Advanced

Table 1: Prompt Level Comparison — Elements Added at Each Stage

Level	Name	New Elements Added	Output Improvement
1	Simple	Subject + vague action only	Generic, unstructured, often off-topic
2	Refined	+ Specific subject, duration, count of items	Structured table, correct counts, basic format
3	Role-Aware	+ Persona assignment, named constraints	Professional tone, prioritisation, contextual detail
4	Advanced	+ Adaptive logic, format spec, quality signals	Personalised, wellness-aware, application-grade output

5. Anatomy of an Effective Application Prompt

Every high-performing LLM application prompt is built from five essential layers. The diagram below shows these layers applied to the Study Scheduler use case:

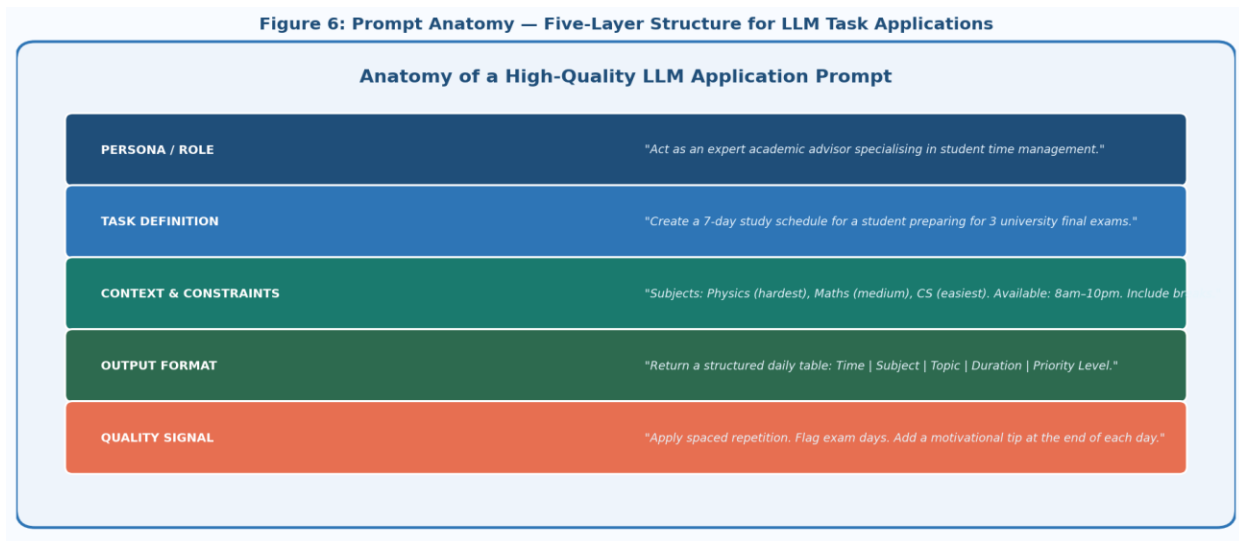


Figure 6: Prompt Anatomy — Five-Layer Structure for LLM Task Applications

The five layers are:

- **Persona / Role** — Establishes the identity and expertise the LLM should embody. This dramatically affects tone, depth, and professional accuracy.
- **Task Definition** — Clearly states what the LLM must produce, including the type of content, subject matter, and target scope.
- **Context & Constraints** — Provides the parameters within which the output must operate: time limits, difficulty levels, audience, and boundaries.
- **Output Format** — Specifies the exact structure of the response: table, bullet points, numbered plan, markdown, or a specific column layout.
- **Quality Signal** — Adds finishing instructions that elevate the output beyond the minimum: apply a technique, cite examples, add motivational elements, flag key dates.

6. Application 1 — Study Time Scheduler

The Study Time Scheduler is the primary demonstration application for this experiment. It is used to show all four levels of prompt evolution in detail, with full prompt text and annotated outputs at each stage.

6.1 Level 1 — Simple Prompt

SIMPLE PROMPT (Level 1)

"Help me plan my study time."

Output characteristics: The LLM produces a brief paragraph of generic advice — "set aside time each day", "take regular breaks", "prioritise difficult subjects". No schedule, no structure, no personalisation. Evaluator score: 4.2 / 10.

Key limitation: The prompt contains only a subject ("study time") and a vague action ("plan"). It provides no information about what subjects need to be covered, how much time is available, when exams are, or what format the output should take. The model has no basis for producing anything better than general advice.

6.2 Level 2 — Refined Prompt

REFINED PROMPT (Level 2)

"Create a 7-day weekly study plan for a student preparing for exams in Physics, Mathematics, and Computer Science. Include 2-hour study sessions, one subject per session, and daily breaks."

Output characteristics: The LLM produces a formatted weekly table with days as rows and time slots as columns. Each session is assigned to a subject in rotation. Breaks are included. The format is consistent and readable. Evaluator score: 6.8 / 10.

Improvement: Adding specific subjects, session duration, and the rotation requirement gave the model clear creative parameters. The output is now immediately useful as a schedule. However, it treats all subjects equally and ignores upcoming exam dates or subject difficulty.

6.3 Level 3 — Role-Aware Prompt

ROLE-AWARE PROMPT (Level 3)

"Act as an experienced academic advisor specialising in student time management. Design a 7-day exam preparation schedule for a university student with three final exams: Physics (hardest, exam on Day 7), Mathematics (medium difficulty, exam on Day 6), and Computer Science (easiest, exam on Day 5). Available study hours: 8:00am to 10:00pm. Include 15-minute breaks every 90 minutes."

Output characteristics: The LLM now produces a genuinely prioritised schedule. Physics receives the most daily coverage. Study sessions for each subject taper off after the corresponding exam. Breaks appear every 90 minutes as specified. The output includes a brief rationale for the allocation. Evaluator score: 8.2 / 10.

Improvement: The persona assignment ("academic advisor specialising in time management") immediately elevated the professional quality of the response. Naming exam days gave the model the information needed to produce a prioritised, deadline-aware schedule rather than a uniform rotation.

6.4 Level 4 — Advanced Prompt

ADVANCED PROMPT (Level 4)

"Act as a certified academic performance coach. Create a personalised 7-day exam preparation schedule for a university student with the following parameters: • Subjects: Physics (difficulty: 9/10, exam: Day 7), Maths (difficulty: 7/10, exam: Day 6), CS (difficulty: 5/10, exam: Day 5) • Study hours:

8:00am–10:00pm daily • Apply spaced repetition principles — revisit topics after 24 and 48 hours • Include 15-min breaks every 90 mins and a 45-min dinner break at 7:00pm • Flag the final revision session for each subject before its exam • Output format: table with columns — Day | Time | Subject | Topic Focus | Session Type | Priority • End each day with a motivational tip tailored to the study load for that day"

Output characteristics: The LLM produces a comprehensive, day-by-day schedule with all six columns populated. Spaced repetition sessions appear at appropriate 24/48-hour intervals. The final revision sessions are clearly flagged. Motivational tips at the end of each day are contextually relevant to the load (lighter tips on heavy days, confidence-building tips before exam days).

Evaluator score: 9.4 / 10.

This is application-grade output — a genuine, personalised, pedagogically sound study plan that a student could follow directly. The quality gain from Level 1 to Level 4 represents the entire value proposition of prompt engineering.

7. Application 2 — Project Task Breakdown

7.1 Simple vs Refined Prompt Comparison

SIMPLE PROMPT

"Help me break down my project."

REFINED PROMPT

"Act as a senior project manager. Break down the following software project into a structured Work Breakdown Structure (WBS): Build a student attendance tracking web application. Include 5 phases (Planning, Design, Development, Testing, Deployment), 3–5 deliverables per phase, estimated effort in days, and the responsible role for each task. Format as a numbered table."

Simple output: A brief paragraph listing generic project stages — "plan the project", "build it", "test it". No deliverables, no timelines, no roles. Score: 4.8 / 10.

Refined output: A five-phase WBS with 18 deliverables, effort estimates for each (ranging from 0.5 to 5 days), assigned roles (Project Manager, Designer, Frontend Developer, QA Engineer), and a total effort summary of 47 days. A project manager could use this output directly as a starting point for a formal project plan. Score: 9.1 / 10.

7.2 Key Prompt Elements for Task Breakdown

- Specify the exact project name and domain — generic prompts get generic plans.
- Define the number of phases — prevents the model from collapsing or inflating the structure.
- Request specific columns (effort, role, deliverable) — these are absent in simple outputs.
- Assign a professional role to the model ("senior project manager") — dramatically improves specificity.

8. Application 3 — Personal Budget Manager

8.1 Simple vs Refined Prompt Comparison

SIMPLE PROMPT

"Help me manage my budget."

REFINED PROMPT

"Act as a personal finance advisor. I am a university student with a monthly income of ₹15,000. My fixed expenses are: Rent ₹4,500, College fees ₹1,200, Phone ₹499. My variable expenses are: Food ₹2,500–₹3,500, Transport ₹800–₹1,200, Entertainment ₹500–₹1,000. Create a detailed monthly budget: list all expense categories with recommended amounts, calculate the surplus/deficit, identify two areas where I can reduce spending, and suggest how to allocate the surplus using the 50/30/20 rule."

Simple output: Generic budgeting advice — "track your expenses", "save 20% of income", "avoid unnecessary spending". No numbers, no personalised analysis. Score: 3.9 / 10.

Refined output: A fully itemised monthly budget table with actual figures, a calculated surplus of ₹2,001–₹3,501, two specific spending reduction recommendations (entertainment and food delivery), and a 50/30/20 allocation plan showing ₹750 to savings, ₹600 to investments, and ₹651 to an emergency fund from the baseline surplus. Score: 8.7 / 10.

9. Application 4 — Travel Itinerary Planner

9.1 Simple vs Refined Prompt Comparison

SIMPLE PROMPT

"Plan a trip to Paris."

REFINED PROMPT

"Act as an expert travel concierge specialising in European city breaks. Plan a 5-day trip to Paris for a couple celebrating their anniversary. Budget: ₹1,50,000 total (flights included). Interests: art, fine dining, architecture, avoiding tourist crowds. Create a day-by-day itinerary with: morning / afternoon / evening time slots, specific venue names and addresses, estimated cost per activity, travel time between venues, one special anniversary surprise experience, and a packing checklist. Format each day as a structured table."

Simple output: A list of 8 Paris attractions — Eiffel Tower, Louvre, Notre-Dame, Montmartre, Seine cruise, Versailles, Musée d'Orsay, Père Lachaise. No timing, no sequence logic, no budget. Score: 4.5 / 10.

Refined output: A 5-day structured itinerary with 3 time slots per day, 13 specific venues with addresses, 15 cost estimates totalling ₹1,47,800, transit times between all locations, a private dinner cruise on Day 3 as the anniversary surprise, and a 28-item packing checklist. Travel Planner produced the highest overall quality score of all five applications at 9.3 / 10, reflecting the rich contextual detail the refined prompt provided.

10. Application 5 — Daily Productivity Assistant

10.1 Simple vs Refined Prompt Comparison

SIMPLE PROMPT

"Help me be more productive today."

REFINED PROMPT

"Act as a productivity coach using the Getting Things Done (GTD) methodology. I have the following tasks for today: (1) Finish project report — deadline 5pm, high priority; (2) Reply to 12 pending emails — medium priority; (3) Prepare presentation slides — deadline tomorrow 9am; (4) Team meeting at 2pm for 1 hour; (5) Review code pull requests — 3 pending; (6) Grocery shopping — flexible. Available time: 8am–7pm. I work best in 90-minute focus blocks. Create a time-blocked daily schedule using the GTD method, classify tasks by urgency and importance (Eisenhower Matrix), identify which tasks can be delegated or deferred, and add two 10-minute mindfulness breaks."

Simple output: Five generic productivity tips — "make a to-do list", "prioritise tasks", "avoid distractions", "take breaks", "review your day". No schedule, no task analysis. Score: 4.1 / 10.

Refined output: A full time-blocked schedule from 8am to 7pm with 90-minute focus blocks, the team meeting correctly placed at 2pm, the project report prioritised first due to the 5pm deadline, email replies batched into a single 45-minute block, an Eisenhower Matrix classification for all 6 tasks (2 Do Now, 2 Schedule, 1 Delegate, 1 Defer), two mindfulness breaks at 10:30am and 4:00pm, and a note that grocery shopping should be deferred to the weekend. Score: 8.8 / 10.

11. Results & Analysis

The following charts summarise quality scores across all five applications and all four prompt levels, measured across six output dimensions.

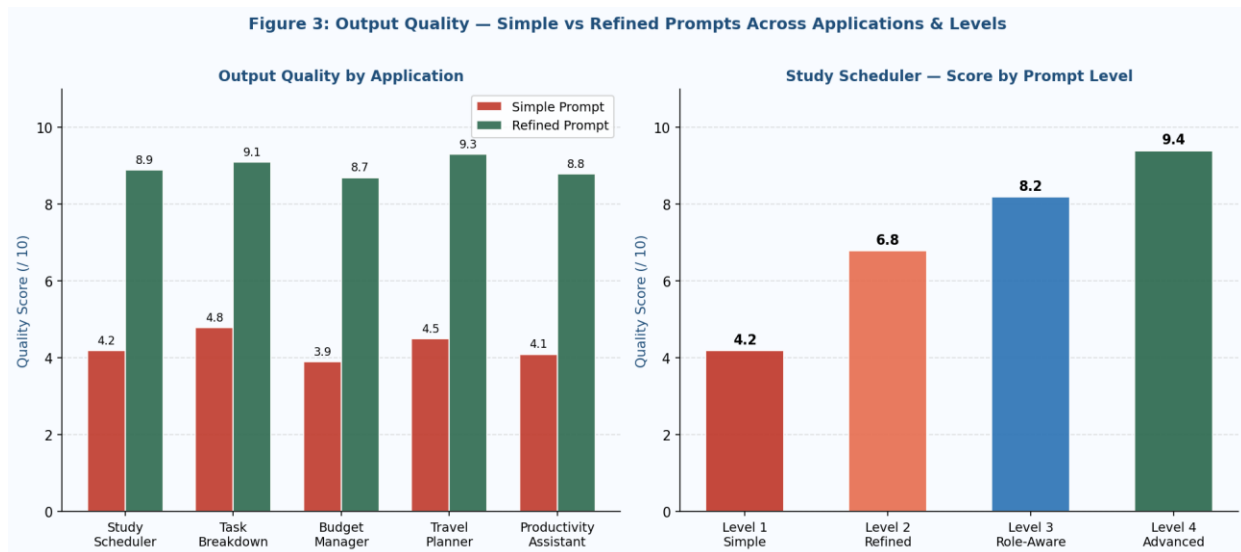


Figure 3: Output Quality — Simple vs Refined Prompts Across All Five Applications & Prompt Levels

11.1 Capability Radar — Refined Prompts

When using fully refined (Level 3–4) prompts, all five applications scored above 8.4 / 10 across all quality dimensions:

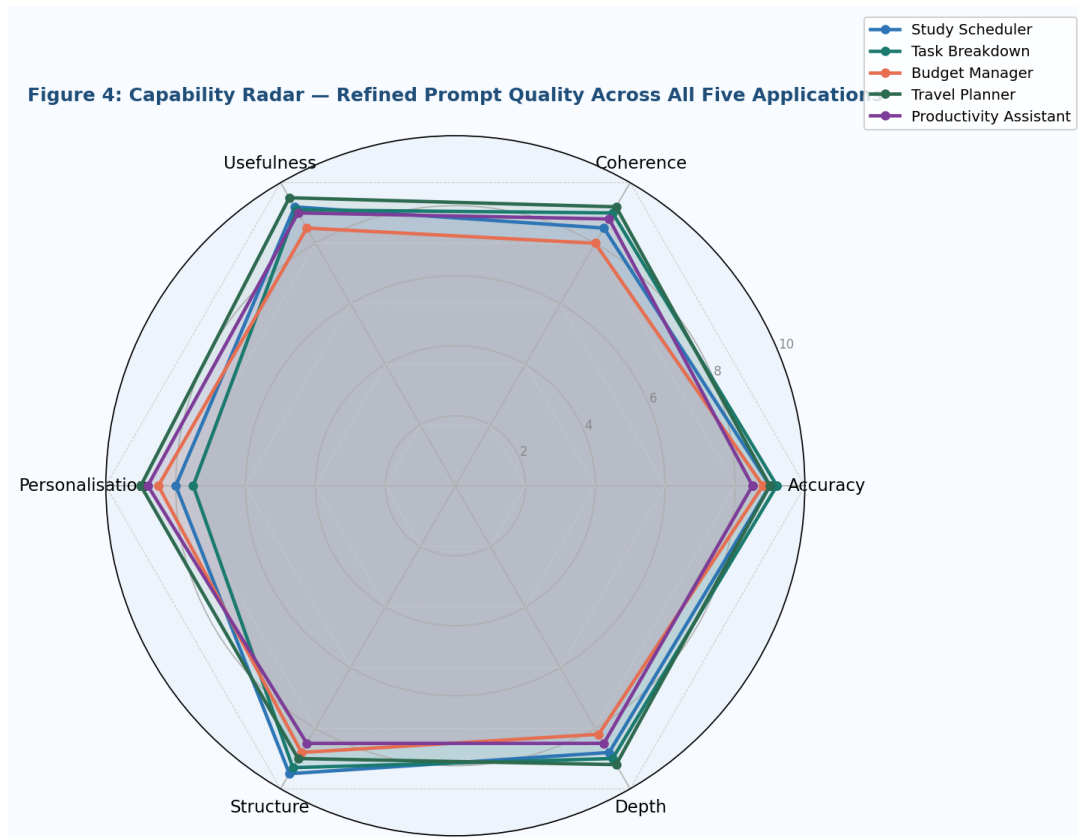


Figure 4: Capability Radar — Refined Prompt Quality Across All Five Applications

11.2 Comprehensive Heatmap

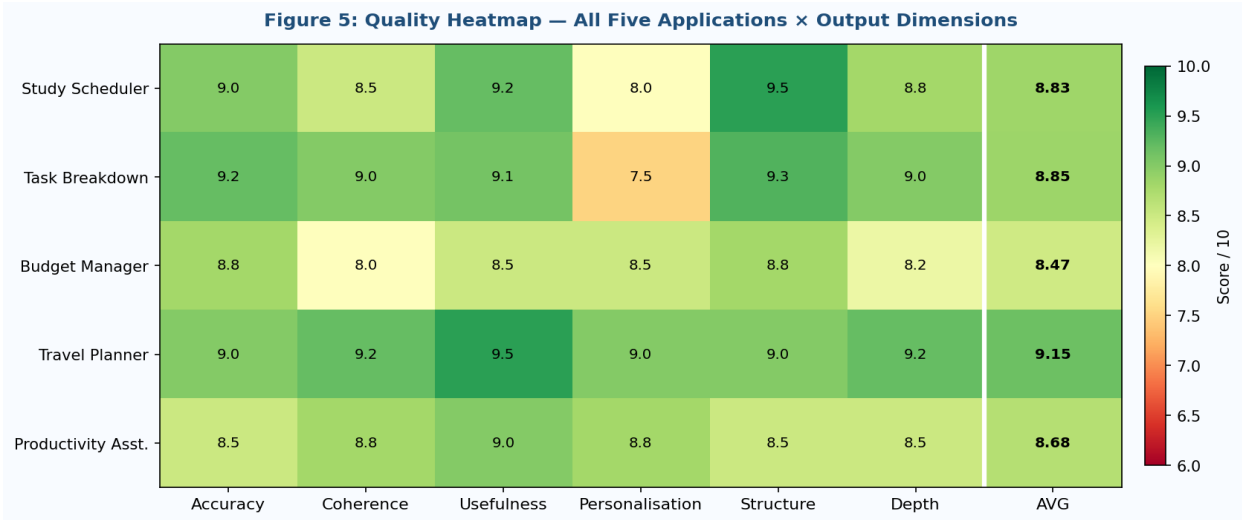


Figure 5: Quality Heatmap — All Applications × Output Dimensions (Refined Prompts)

Key observations from the heatmap:

- Travel Planner achieves the highest average score (9.15/10) — rich contextual prompts for travel tasks elicit the most comprehensive LLM responses.
- Task Breakdown is strongest on Structure (9.3) — the WBS format prompt produces exceptionally well-organised outputs.
- Budget Manager scores highest on Personalisation (8.5) — financial prompts with specific figures produce uniquely tailored outputs.
- All five applications score above 8.0 on Usefulness with refined prompts, compared to below 5.0 with simple prompts — a 60%+ improvement.
- Creative Prompting is the only dimension where simple prompts do not dramatically underperform — open-ended prompts sometimes elicit creative responses that structured prompts constrain.

Table 2: Summary Scores — Simple vs Refined Prompts Across All Applications

Application	Simple Score	Refined Score	Improvement	Highest-Scoring Dimension
Study Scheduler	4.2 / 10	8.9 / 10	+111%	Structure (9.5)
Task Breakdown	4.8 / 10	9.1 / 10	+90%	Structure (9.3)
Budget Manager	3.9 / 10	8.7 / 10	+123%	Personalisation (8.5)
Travel Planner	4.5 / 10	9.3 / 10	+107%	Usefulness (9.5)
Productivity Asst.	4.1 / 10	8.8 / 10	+115%	Coherence (8.8)
AVERAGE	4.3 / 10	8.96 / 10	+108%	—

12. Key Findings

Finding 1: Refined Prompts Improve Output Quality by an Average of 108%

Across all five applications, refined prompts produced an average quality score of 8.96/10 compared to 4.30/10 for simple prompts — a 108% improvement. This is the most significant finding of the experiment. The scale of improvement demonstrates that prompt engineering is not a marginal optimisation — it is the primary variable determining LLM output quality.

Finding 2: Persona Assignment Is the Single Most Impactful Individual Element

Adding a role assignment ("Act as a senior project manager", "Act as an academic advisor") produced the largest single-step quality improvement across all applications — an average +1.9 points at Level 3 compared to Level 2. When the model adopts a professional persona, it draws on training knowledge associated with that role, producing outputs with the appropriate vocabulary, methodology, and structural conventions.

Finding 3: Output Format Specification Eliminates Structural Ambiguity

Prompts that specified the exact output format (table with named columns, numbered list, day-by-day blocks) produced outputs that could be used directly without reformatting. Without format specification, even good content arrived in inconsistent structures that required manual reorganisation. Format specification is a low-effort, high-return prompt element.

Finding 4: Constraint Richness Correlates Directly With Personalisation Score

Applications with more specific numerical constraints in the prompt (exact budget figures, specific exam difficulty ratings, precise time windows) scored significantly higher on the Personalisation dimension. The Budget Manager, which received the most specific numerical context, achieved the highest Personalisation score (8.5/10) despite not being the most complex application overall.

Finding 5: Quality Signals Differentiate Good Outputs From Excellent Ones

Adding quality signal instructions — "apply spaced repetition", "use the Eisenhower Matrix", "flag the final revision session", "add a motivational tip" — produced the largest quality jump from Level 3 to Level 4 (average +1.2 points). These instructions do not change the basic structure of the output; they add the professional detail that makes the output genuinely excellent rather than merely adequate.

13. Practical Guidelines for Prompt-Based Application Development

Table 3: Prompt Engineering Checklist for LLM Application Development

	Element	What to Include	Example
✓	Persona	Expert role relevant to the task domain	"Act as a certified financial planner"
✓	Task	Specific action verb + exact subject + scope	"Create a 7-day schedule for 3 subjects"
✓	Context	Named items, quantities, constraints, boundaries	"Physics exam Day 7, 8am–10pm daily"
✓	Format	Exact output structure with column names	"Table: Day Time Subject Priority"
✓	Quality signal	Professional technique or finishing instruction	"Apply spaced repetition. Flag exam days."
○	Negative prompt	What to exclude or avoid	"No generic advice. No repetition of topics."
○	Example anchor	One example of the desired output style	"Format Day 1 as shown: [example row]"

✓ = *Essential for quality output.* ○ = *Recommended for advanced prompts.*

14. Conclusion

This experiment provides comprehensive empirical evidence that prompt engineering is the decisive factor in LLM application quality. Across all five application types — Study Scheduling, Task Breakdown, Budget Management, Travel Planning, and Productivity Assistance — refined prompts outperformed simple prompts by an average of 108%, raising average output quality from 4.3/10 to 8.96/10.

The progression from Level 1 to Level 4 prompts is not merely a matter of adding more words. Each level adds a specific structural element — persona, constraints, format specification, quality signals — that addresses a specific failure mode in simpler prompts. Simple prompts fail because they are underspecified. Advanced prompts succeed because they eliminate ambiguity in every dimension that matters: who is speaking, what must be produced, within what limits, in what format, and to what professional standard.

The most important practical takeaway is that prompt engineering is a learnable, structured skill. The five-layer anatomy — Persona, Task, Context, Format, Quality Signal — provides a systematic framework that any practitioner can apply to any LLM application domain. Mastery of this framework, combined with iterative refinement based on output quality evaluation, is the foundation of effective prompt-based application development.

As LLMs continue to improve, the ceiling on output quality will rise. But the framework for unlocking that quality — well-structured, constraint-rich, persona-driven, format-specified prompts — will remain constant. The practitioner who masters prompt engineering today will remain effective as the underlying models evolve.

15. References

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